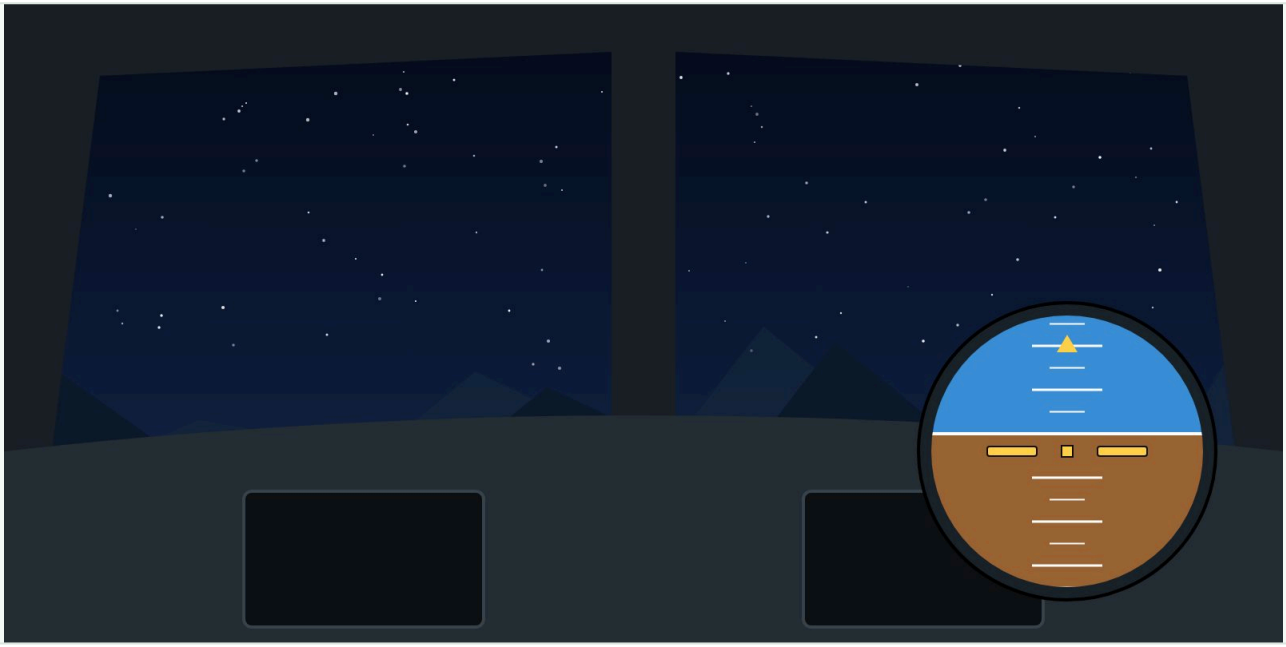
MODULE 2 · EGPWS AND THE ESCAPE MANOEUVRE

Lesson 05: EGPWS / TAWS: Alerts and Crew Response

Lesson 5 of 7 · about 8 min

LEARNING OBJECTIVES

- Describe the EGPWS basic modes and enhanced functions.
- Distinguish cautions from warnings.
- Apply the company response to EGPWS cautions.



At night or in IMC, follow an EGPWS warning immediately.

The system

A terrain awareness warning system (TAWS) is required for commercial aeroplanes (ICAO Annex 6; EASA CAT.IDE.A.150). Our fleets use the **Enhanced GPWS (EGPWS)**, which combines the classic ground proximity modes with a **terrain database** and the aircraft position.

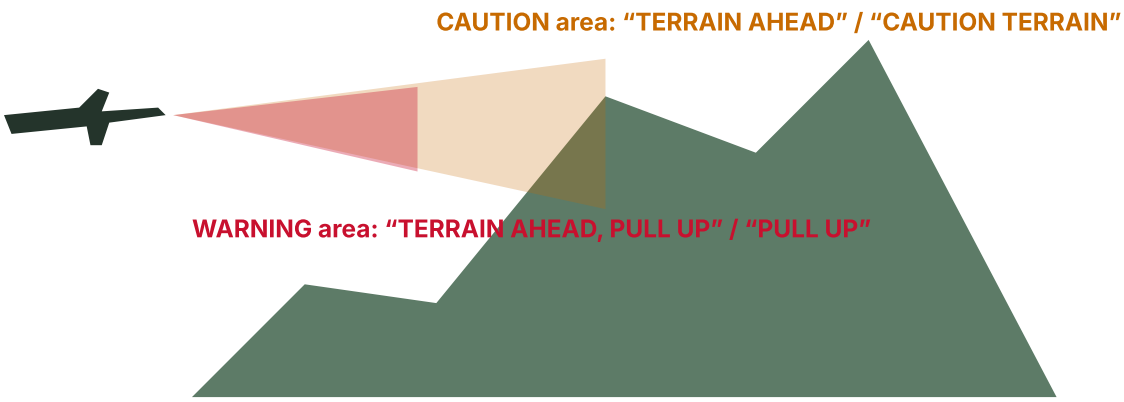
Mode	Protects against	Typical alerts
1	Excessive descent rate	“SINK RATE”, “PULL UP”
2	Excessive terrain closure rate	“TERRAIN TERRAIN”, “PULL UP”
3	Altitude loss after take-off or go-around	“DON'T SINK”
4	Unsafe terrain clearance when not in landing configuration	“TOO LOW GEAR”, “TOO LOW FLAPS”, “TOO LOW TERRAIN”
5	Excessive deviation below the glideslope	“GLIDESLOPE”
6	Advisory callouts	Altitude callouts, “BANK ANGLE”

Enhanced	Terrain and obstacles ahead (look-ahead); terrain clearance floor near runways	“CAUTION TERRAIN”, “CAUTION OBSTACLE”, “TERRAIN AHEAD”, “TERRAIN AHEAD PULL UP”, “OBSTACLE PULL UP”, “TOO LOW TERRAIN”

Alert wording varies with the installation — check the aircraft manuals.

Enhanced function: look-ahead against the terrain database

The caution comes first (typically up to about a minute ahead), the warning later (typically about 30 second



The look-ahead function depends on accurate aircraft position and an up-to-date terrain database. Exact timings depend on the installation.

Cautions and warnings

- **Caution** (amber) — for example “SINK RATE”, “TERRAIN TERRAIN”, “DON'T SINK”, “TOO LOW FLAPS”, “TOO LOW GEAR”, “CAUTION TERRAIN”, “CAUTION OBSTACLE”, “BANK ANGLE”, “GLIDESLOPE”. **Company SOP**: the crew shall immediately assess the flight path or configuration with respect to the ground, and the PF shall take appropriate action to correct the situation. Q200 SOP: the PF verbalises the action to the PM with **“CORRECTING”**.
- **Warning** (red) — for example “TOO LOW TERRAIN”, “TERRAIN PULL UP”, “OBSTACLE PULL UP” or “PULL UP” (“WHOOP WHOOP PULL UP”). **Apply the escape manoeuvre immediately** (Lesson 6).

NO DELAY

At night or in IMC, a warning must be followed **without hesitation and without trying to confirm it visually**. The Q200 SOP adds: “The flight crew must use a good judgment in this matter.” A warning should only ever be judged spurious in daylight VMC, with the terrain and obstacles clearly in sight and no doubt about the aircraft's position. Report every alert.

KEY TAKEAWAYS

- EGPWS = basic modes 1–6 plus terrain database look-ahead.
- Caution: assess the flight path and correct (“CORRECTING”).
- Warning (“PULL UP”, “TERRAIN AHEAD PULL UP”): escape manoeuvre immediately.
- At night or in IMC, never wait to confirm a warning visually.

KNOWLEDGE CHECK

At night in IMC you hear “TERRAIN AHEAD, PULL UP”. What do you do?

A. Check the chart first

B. Apply the escape manoeuvre immediately

C. Reduce speed and level off

D. Ask ATC to confirm terrain clearance

Answer: B. A warning at night or in IMC requires the escape manoeuvre immediately, without trying to confirm it.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.